

# Solve long multiplication problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: A theatre has 1,248 seats in each of 24 identical sections. How many seats are there? Copy or complete the first step.

2. A factory packs 2,315 items each day for 32 days. How many items?

3. A school orders 48 boxes of 1,275 pencils. How many pencils?

4. Miro says: Miro calculates correctly but gives a product with no unit or context. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: A theatre has 1,248 seats in each of 24 identical sections. How many seats are there? Copy or complete the first step.

**Answer** 29,952 seats.

2. A factory packs 2,315 items each day for 32 days. How many items?

**Answer** 74,080 items.

3. A school orders 48 boxes of 1,275 pencils. How many pencils?

**Answer** 61,200 pencils.

4. Miro says: Miro calculates correctly but gives a product with no unit or context. Is this correct? Explain.

**Answer** Return to the question, label the answer and complete every step in the context.

# Solve long multiplication problems

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. A theatre has 1,248 seats in each of 24 identical sections. How many seats are there?

6. Check this result using an inverse, estimate or known fact: A school orders 48 boxes of 1,275 pencils. How many pencils?

2. A factory packs 2,315 items each day for 32 days. How many items?

7. The school gives 18 pencils to each of 3,200 pupils. How many remain from 61,200?

3. A school orders 48 boxes of 1,275 pencils. How many pencils?

8. Identify and correct this misconception: Miro calculates correctly but gives a product with no unit or context.

4. Solve this independently and show every stage: A theatre has 1,248 seats in each of 24 identical sections. How many seats are there?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A factory packs 2,315 items each day for 32 days. How many items?

10. Write and solve a similar question to: A school orders 48 boxes of 1,275 pencils. How many pencils?

# Teacher answers and checking prompts

1. A theatre has 1,248 seats in each of 24 identical sections. How many seats are there?  
**Answer** 29,952 seats.
2. A factory packs 2,315 items each day for 32 days. How many items?  
**Answer** 74,080 items.
3. A school orders 48 boxes of 1,275 pencils. How many pencils?  
**Answer** 61,200 pencils.
4. Solve this independently and show every stage: A theatre has 1,248 seats in each of 24 identical sections. How many seats are there?  
**Answer** 29,952 seats.
5. Use a second method or representation for: A factory packs 2,315 items each day for 32 days. How many items?  
**Answer** Expected result: 74,080 items. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A school orders 48 boxes of 1,275 pencils. How many pencils?  
**Answer** Expected result: 61,200 pencils. A valid check must be shown.
7. The school gives 18 pencils to each of 3,200 pupils. How many remain from 61,200?  
**Answer** 3,600 pencils.
8. Identify and correct this misconception: Miro calculates correctly but gives a product with no unit or context.  
**Answer** Return to the question, label the answer and complete every step in the context.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A school orders 48 boxes of 1,275 pencils. How many pencils?  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Solve long multiplication problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. The school gives 18 pencils to each of 3,200 pupils. How many remain from 61,200?

6. Create a new example that tests the same idea as: A school orders 48 boxes of 1,275 pencils. How many pencils?

2. Prove the answer to this question, rather than only calculating it: A theatre has 1,248 seats in each of 24 identical sections. How many seats are there?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A factory packs 2,315 items each day for 32 days. How many items?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 61,200 pencils.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro calculates correctly but gives a product with no unit or context.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. The school gives 18 pencils to each of 3,200 pupils. How many remain from 61,200?  
**Answer** 3,600 pencils.
2. Prove the answer to this question, rather than only calculating it: A theatre has 1,248 seats in each of 24 identical sections. How many seats are there?  
**Answer** Expected result: 29,952 seats. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A factory packs 2,315 items each day for 32 days. How many items?  
**Answer** Expected result: 74,080 items. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 61,200 pencils.  
**Answer** One valid reconstruction is: A school orders 48 boxes of 1,275 pencils. How many pencils? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro calculates correctly but gives a product with no unit or context.  
**Answer** Return to the question, label the answer and complete every step in the context.
6. Create a new example that tests the same idea as: A school orders 48 boxes of 1,275 pencils. How many pencils?  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.